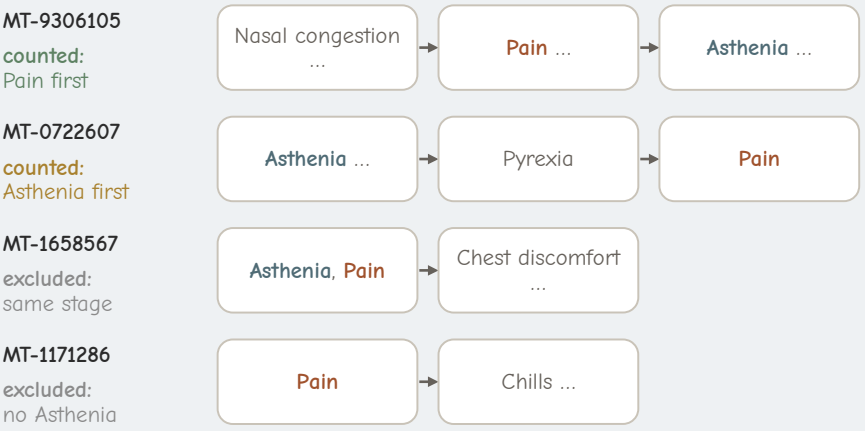
Claim (P-V1) After Moderna, Pain typically occurs before Asthenia.
gold: **supported** (k = 18 of n = 22, p = 0.0022)

(a) Which reports can witness the order?



Stages run left to right; events at one stage co-exist.

Moderna: 983 reports, 30 with both events; **8 at the same stage are excluded**, so n = 22 and k = 18 (Pain first).

(b) EHRSQL-C: one query

```
in_denominator = Pain.stage < Asthenia.stage
in_numerator   = Pain.stage < Asthenia.stage
```

n = 18, k = 18 → $p = 0.5^{18} \approx 4 \times 10^{-6}$ → **supported**

Right verdict, wrong evidence: the 4 Asthenia-first reports are dropped, so n = k and the claim is supported whatever the data say.

(c) ART: articulate, retrieve, test

S1 Articulate

denominator: both events at **distinct** stages; null $p_0 = 0.5$; one-sided exact binomial; $n \geq 10$

S2 Retrieve

```
in_denominator = Pain.stage ≠ Asthenia.stage
in_numerator   = Pain.stage < Asthenia.stage
```

S3 Test

n = 22, k = 18 (the gold sets); $P(X \geq 18) = 0.0022$ → **supported**

Right verdict on the right evidence: every counted report matches the gold set.